



RAMCO INSTITUTE OF TECHNOLOGY

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Department of Computer Science and Engineering

Academic Year 2025– 2026 (Odd Semester)

Degree, Semester & Branch : V Semester B.E CSE
Course Code & Title : CCS366 & Software Testing and Automation
Name of the Faculty member : Mrs. S. Vijaya Amala Devi, AP / CSE

Innovative Practice Description

- **Unit / Topic:** Unit III/Test Design and Execution/Test Design Factor
- **Course Outcome:**3
- **Topic Learning Outcome:** 3b
- **Activity Chosen:** Peer Learning
- **Justification:**
 - “Test Design Factors” is an ideal topic for peer learning because it requires analytical thinking, comparison of ideas, and collaborative decision-making. Students must evaluate requirements, risks, constraints, and coverage criteria to decide how test cases should be designed.
 - Discussing these factors in groups helps learners understand multiple viewpoints, deepen their reasoning skills, and apply theoretical concepts to practical testing situations. This active exchange of ideas ensures better clarity and retention than passive classroom learning.
 - Peer learning also mirrors real-world software testing environments, where testers work together to analyse scenarios, prioritize test conditions, and justify their testing strategies. By explaining their decisions to peers, students enhance their communication skills, teamwork abilities, and problem-solving capabilities.
 - This collaborative approach enables students to gain practical experience in designing effective test cases and prepares them for professional roles in the software industry.
- **Time Allotted for the Activity:** 20 Minutes
- **Details of the Implementation:**
 - The instructor began the session with a brief overview of *software testing fundamentals*, including the purpose of testing, types of test cases, and the importance of quality assurance. This introduction ensured that all students had the foundational understanding needed before moving into the peer teaching activity.
 - **Figure 1:** Shows a student group reviewing basic testing concepts, demonstrating initial engagement and preparation for the peer learning session.
 - Based on students’ interest and readiness, **Ms. Kavyalskshmi** was selected as the peer teacher for the session.
 - The peer teacher began by introducing the major **Test Design Factors** and explained how each factor affects the structure, priority, and quality of test cases in software testing. The concepts covered included:

- **Requirement Analysis** – understanding functional and non-functional requirements before creating test cases.
- **Risk Assessment** – identifying high-risk modules that require more detailed or prioritized testing.
- **Test Data Availability** – explaining how data completeness and correctness influence test effectiveness.
- **Time & Resource Constraints** – discussing how limited testing time or tools may affect test depth and coverage.
- **Test Coverage Criteria** – describing equivalence partitioning, boundary value analysis, and decision tables as techniques to maximize coverage.
- The peer teacher conducted a set of **real-time demonstrations** using sample test scenarios and case studies:
 - Presented a requirement document and showed how to derive test conditions.
 - Demonstrated how risk levels change the priority of test cases.
 - Illustrated test data preparation using positive, negative, and boundary data examples.
 - Showed how constraints like deadlines and team size affect test planning.
 - Used flowcharts and tables to demonstrate how coverage techniques are applied.
- Students were actively engaged throughout the session. The peer teacher interacted with classmates by asking them to predict how changes in factors—such as missing test data, unclear requirements, or limited time—would impact test quality. Teams collaboratively discussed practical applications, such as how e-commerce apps, banking systems, or hospital management systems would require different testing strategies depending on risk and requirement complexity.
- The instructor concluded the session by appreciating the peer teacher's effort, offering feedback, and summarizing important takeaways: the importance of test design factors in creating meaningful, efficient, and high-quality test cases. The peer learning activity improved students' understanding, analytical skills, and collaborative problem-solving abilities, making complex testing concepts easier to grasp and apply in real-world scenarios.

CO – PO / PSO mapping:

CO	PO1	PO2	PO5	PO8	PO10	PO12	PSO3
CO4	3	3	3	3	3	3	3

(1 – Low 2 – Moderate 3 – High)

• **PO / PSO mapped:**

Innovative practice	PO1	PO2	PO4	PO5	PO10	PO12	PSO2
	3	3	3	1	2	2	2
Justification for correlation	Student Recognizing effective test	Student Effective test case identif	Student Identifying effective test cases	Student Leveraging modern testing tools and techniques	Student Communicating the importance of identifying	Student to Recognizing the importance of identifying effective test	Students can Identifying effective test cases involves considering IoT, cloud

	cases requires a deep understanding of software engineering principles to identify critical functionalities and potential failure points.	ication involves analyzing application requirements and potential scenarios to pinpoint areas susceptible to critical defects.	requires the ability to investigate complex application behaviors and dependencies to uncover potential critical defects.	aids in identifying effective test cases by enhancing test coverage and efficiency in defect detection.	effective test cases to stakeholders and team members ensures alignment and support for testing efforts.	cases requires a commitment to continuous learning to stay updated with evolving testing methodologies and technologies.	computing, and cybersecurity aspects to uncover critical defects, contributing to the reliability of IT solutions
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• **Images / Screenshot of the practice:**



Fig 1: Student discussed the topics

Reflective Critique:

❖ **Feedback from Students and Stakeholders:**

- Students appreciated the opportunity to explore **test design factors** independently before the peer teaching session.
- Access to pre-class materials allowed them to review key concepts such as **requirement analysis, risk assessment, test data preparation, coverage criteria, and resource constraints** at their own pace.

- During the session, discussing these topics with peers helped them **clarify doubts, strengthen conceptual understanding, and apply theoretical testing principles to practical scenarios**, including designing test cases, prioritizing based on risk, and evaluating test coverage for real-world applications.

❖ **Benefits of the Practice:**

- Peer teaching encouraged students to actively communicate and work together. Group discussions and team-based explanations fostered collaboration, helping learners clarify doubts, exchange ideas, and support one another's understanding of complex concepts related to test design factors, such as requirement analysis, risk prioritization, and coverage techniques.
- By teaching and discussing topics such as requirement analysis, risk assessment, test data preparation, coverage criteria, and resource constraints, students developed a deeper conceptual understanding. Explaining testing concepts to peers reinforced their own knowledge and made theoretical ideas more concrete, improving retention and confidence in designing effective test cases.
- The use of sample requirements, risk charts, test data tables, flow diagrams, and real-world testing scenarios made learning more interactive and practical. Students were able to relate theoretical testing principles to real-world software applications—such as e-commerce platforms, banking systems, and mobile apps—enhancing their motivation and helping them understand how test design factors shape testing quality in industry settings.

❖ **Challenges Faced:**

- Not all students were willing to take on the role of peer teacher, which limited the number of active presentations during the session. This made it challenging to ensure equal opportunities for students to lead discussions, demonstrate test design techniques, and share their understanding of software testing concepts.
- Some students struggled with understanding or correctly using technical terms related to test design factors, such as *requirement analysis*, *risk assessment*, *boundary value analysis*, *equivalence partitioning*, and *test coverage criteria*. These students required additional guidance from the peer teacher or faculty to fully grasp the concepts, highlighting the need for proper scaffolding, examples, and support during peer-led testing sessions

References:

- ❖ https://en.wikipedia.org/wiki/Peer_learning
- ❖ [Peer to Peer Learning - Examples, Benefits & Strategies](#)
- ❖ <https://whatfix.com/blog/peer-to-peer-learning/>

Signature of Faculty Member

HOD